ES-446 Assignment 2

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Question 1: The Nabla Operator

The **nabla (del) operator** (∇) is a vector differential operator defined in Cartesian coordinates as:

$$\nabla = \mathbf{i}\frac{\partial}{\partial x} + \mathbf{j}\frac{\partial}{\partial y} + \mathbf{k}\frac{\partial}{\partial z}$$

It is a **linear operator** that satisfies additivity and homogeneity. While it possesses vector-like components, it only yields physical results when acting upon a field.

Question 2: Fields and Differentiation

A **field** is a physical quantity assigned to every point in space.

- **Scalar Field:** Defined by magnitude only. *Example:* Altitude of points on a geographical map.
- **Vector Field:** Defined by magnitude and direction. *Example:* The velocity profile of water particles in a flowing river.

Question 3: Vector Calculus Operations

Given $f(x, y, z) = 2x+3xy+2z^2$ **(scalar field)** and $\mathbf{g}(x, y, z) = 2x\mathbf{i}+3xy\mathbf{j}+2z^2\mathbf{k}$ **(vector field)**, we evaluate the following at point $(1, 2, 3)$:

1. Gradient (∇)

- **Field f:** $\nabla f = (2 + 3y)\mathbf{i} + (3x)\mathbf{j} + (4z)\mathbf{k}$.
  At $(1, 2, 3)$, $\nabla f = \mathbf{8i} + \mathbf{3j} + \mathbf{12k}$.
- **Field g:** **Not Defined.** Standard gradient operation is reserved for scalar fields.

2. Divergence ($\nabla\cdot$)

- **Field f: Not Defined.** Divergence requires a vector input and cannot be applied to a scalar field.
- **Field g:** $\nabla \cdot \mathbf{g} = \frac{\partial(2x)}{\partial x} + \frac{\partial(3xy)}{\partial y} + \frac{\partial(2z^2)}{\partial z} = 2 + 3x + 4z$.
  At $(1, 2, 3)$, $\nabla \cdot \mathbf{g} = 2 + 3(1) + 4(3) = \mathbf{17}$.

3. Curl ($\nabla\times$)

- **Field f: Not Defined.** Curl is a cross product operation and is not applicable to scalar fields.
- **Field g:** $\nabla \times \mathbf{g} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x & 3xy & 2z^2 \end{vmatrix} = \mathbf{i}(0) - \mathbf{j}(0) + \mathbf{k}(3y)$.
  At $(1, 2, 3)$, $\nabla \times \mathbf{g} = \mathbf{6k}$.

Question 4: Python Implementation

The following SymPy script was used to verify the manual calculations.

```python
import sympy as sp

x, y, z = sp.symbols('x y z')
f = 2*x + 3*x*y + 2*z**2
g = sp.Matrix([2*x, 3*x*y, 2*z**2])
pt = {x: 1, y: 2, z: 3}

# Gradient of f
grad_f = [sp.diff(f, v).subs(pt) for v in (x, y, z)]

# Divergence of g
div_g = (sp.diff(g[0], x) + sp.diff(g[1], y) + sp.diff(g[2], z)).
    subs(pt)

# Curl of g
curl_g = sp.Matrix([
    sp.diff(g[2], y) - sp.diff(g[1], z),
    sp.diff(g[0], z) - sp.diff(g[2], x),
    sp.diff(g[1], x) - sp.diff(g[0], y)
]).subs(pt)

print(f"Grad f: {grad_f}, Div g: {div_g}, Curl g: {curl_g}")
```